

REVIEW & INTERPRETATION

Shifts in Soybean Yield, Nutrient Uptake, and Nutrient Stoichiometry: A Historical Synthesis–Analysis

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ABSTRACT

Few studies have investigated changes over time in nutrient uptake and yield, in addition to the study of nutrient stoichiometry as a metric of nutrient limitations in soybean [*Glycine max* (L.) Merr.]. A comprehensive synthesis–analysis was performed by compiling a global historical soybean database of yield, total biomass, and nutrient (N, P, and K) content and concentration in studies published from 1922 to 2015. This period was divided in three eras based on genetically modified soybean events: Era I (1922–1996), Era II (1997–2006), and Era III (2007–2015). The main findings of this review are: (i) seed yield improved from 1.3 Mg ha^{−1} in the 1930s to 3.2 Mg ha^{−1} in the 2010s; (ii) yield increase was primarily driven by increase in biomass rather than harvest index (HI); (iii) both N and P HIs increased over time; (iv) seed nutrient concentration remained stable for N and declined for both P (18%) and K (13%); (v) stover nutrient concentration remained stable for N, diminished for P, and increased for K; (vi) nutrient ratios portray different trends for N/P (Era I and III > II), N/K (Era I > II and III), and K/P (Era II and III > I); (vii) yield per unit of nutrient uptake (internal efficiency) increased for N (33%) and P (44%) and decreased for K (11%); and (viii) variations in nutrient internal efficiency were primarily explained by increase in nutrient HI for N and K, but equally explained by both HI for P and seed P concentration. These findings have implications for soybean production and integrated nutrient management to improve yield, nutrient use efficiency, and seed nutrient composition.

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Abbreviations: HI, harvest index; IE, internal efficiency; KHI, potassium harvest index; KIE, potassium internal efficiency; K_{seed}, seed potassium concentration; K_{stover}, stover potassium concentration; NHI, nitrogen harvest index; NIE, nitrogen internal efficiency; N_{seed}, seed nitrogen concentration; N_{stover}, stover nitrogen concentration; PHI, phosphorus harvest index; PIE, phosphorus internal efficiency; P_{seed}, seed phosphorus concentration; P_{stover}, stover phosphorus concentration.

GLOBALLY, soybean [*Glycine max* (L.) Merr.] is the largest source of both animal protein feed and vegetable oil, providing a variety of nutrients and essential elements important for human health (FAO, 2002). Soybean meal, produced in the crushing and oil extraction process, accounts for 65% of protein feed worldwide. Between 1961 and 2014, global soybean production rose 10-fold to reach >306 million Mg, with an average yield increase from 1.1 to 2.6 Mg ha^{−1} (FAO, 2017). For the United States, soybean yield gain from 1922 to 2007 was 25 to 30 kg ha^{−1} yr^{−1} (Specht and Williams, 1984; De Bruin and Pedersen, 2009; Specht et al., 2014). The global yield increase of 1.3% (current for 2013) will not be sufficient to meet the required production by 2050 (Ray et al., 2013).

The causes of soybean yield improvement included changes in environmental conditions, genetic improvement, management practices, and the interactions among these factors. Increases in atmospheric CO₂ and O₃ concentration, air temperature, and climate variability affected yields in the previous decades (Curry et al., 1995; Grashoff et al., 1995; Southworth et al., 2002). Under

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elevated CO₂, total biomass in soybean increased proportionally more than yield, with positive changes in the photosynthetic rate and leaf area, and negative changes for harvest index (HI), stomata conductance, and Rubisco activity (Ainsworth et al., 2002). Kucharik and Serbin (2008) showed that increasing summer temperature could potentially decrease soybean yield in the United States by 16%, whereas increased precipitation might produce a counter effect, improving yield by 5 to 10%.

Traits that contribute to improved soybean yield include longer reproductive or seed-filling periods (Gay et al., 1980; Kumudini et al., 2001; Shen and Liu, 2015), decreased lodging (Specht and Williams, 1984), and improved disease resistance (Foulkes et al., 2009). Changes in management practices that increased soybean yields are related to narrow rows (Heatherly and Elmore, 2004), improvement of weed control (Pike et al., 1991; Osteen, 1993; Bradley and Sweets, 2008), conservation tillage and reduction in harvest losses (Heatherly and Elmore, 2004), and early sowing (Wilcox and Frankenberger, 1987; Conley and Santini, 2007; Bastidas et al., 2008; Sacks and Kucharik, 2011). Early sowing can increase yield by lengthening both the vegetative and seed-filling phases (Egli and Cornelius, 2009).

After historical changes in crop yield, variations in nutrient uptake and related nutrient efficiencies have implications for crop and soil management, breeding, and seed quality, as illustrated for sorghum [*Sorghum bicolor* (L.) Moench; Ciampitti and Prasad, 2016], corn (*Zea mays* L.; Ciampitti and Vyn, 2012, 2014), wheat (*Triticum aestivum* L.; Guttieri et al., 2017), and legumes and oil seed crops (Sadras, 2006). From a historical perspective, few studies analyzed nutrient uptake, partitioning, and remobilization for soybean; those available took place in the 1930s (Borst and Thatcher, 1931), 1950s (Hammond et al., 1951), 1970s (Hanway and Weber, 1971; Harper, 1971), and 2010s (Bender et al., 2015). Bender et al. (2015) reported maximum yields of 3.5 Mg ha⁻¹ in modern soybean varieties without exploring high yielding levels (>6 Mg ha⁻¹). Gaspar et al. (2017) evaluated nutrient uptake for soybean in the US Midwest in a range of yields (3.6–5.4 Mg ha⁻¹). Neither of the abovementioned studies characterized historical trends of nutrient uptake nor nutrient stoichiometry. Furthermore, published studies on soybean plant nutrition have largely focused on individual nutrients, mostly on N (Fabre and Planchon, 2000; Salvagiotti et al., 2008, 2009; Jin et al., 2011; Rotundo et al., 2014; Van Roekel and Purcell, 2014; Divito et al., 2016; Cafaro La Menza et al., 2017), less on P (Pan et al., 2008; Wang et al., 2009, 2010; van de Wiel et al., 2016), and a few on K (Pettigrew, 2008; Parvej et al., 2015). Nutrient stoichiometry is useful to understand crop nutrient status, and nutrient ratios can be useful to comprehend nutrient supplies (Mo et al., 2015). Some recent studies have focused on nutrient stoichiometry in soybean (Mallarino et al., 2011;

Salvagiotti et al., 2012; Divito and Sadras, 2014; Divito et al., 2016; Tamagno et al., 2017). For example, S/N ratio is a better indicator of soybean S status than S concentration alone (Divito et al., 2016). To the best of our knowledge, there is the gap of an historical analysis that identifies possible shifts in yield and nutrient-related traits that can help to understand soybean plant nutrition. A historical database of soybean yield and uptake and partitioning of N, P, and K was compiled to characterize historical shifts in these traits and dissect the main physiological components related to nutrient use efficiency, seed nutrient composition, and nutrient (N, P, and K) stoichiometry.

MATERIALS AND METHODS

Approach

We pursued a synthesis-analysis, as in previous works (Ciampitti and Vyn, 2012, 2013, 2014; Ciampitti and Prasad, 2016; Tamagno et al., 2017). There is a tradeoff in pursuing a synthesis analysis rather than a meta-analysis. A synthesis-analysis aggregates a large amount of data and summarizes trends but does not provide a quantitative measure of the effect size (Ciampitti and Vyn, 2012). A meta-analysis can calculate the effect size but requires measures of variation (individual replications, standard deviation) that is not reported by treatment in most of the references collected for this study (Curtis and Wang, 1998). Usually this information is not available, restricting the number of datasets that can be included in an analysis.

Data Search Criteria

We focused on time trends in soybean yield, nutrient uptake and nutrient partitioning, nutrient internal efficiency (i.e., yield per unit nutrient uptake), and nutrient stoichiometry. Papers were retrieved from CABI, Web of Science Core Collection, Scopus, Springer Link, Agricola, and Google Scholar using the keywords: “soybean,” “nitrogen,” “phosphorous,” “potassium,” “seed yield,” “nutrient uptake,” “nutrient ratio,” “harvest index,” and “internal efficiency.” In addition, unpublished data available were included. Only field experiments were included in the database, including those that studied (i) seed yield; (ii) aboveground biomass (total biomass) at maturity (end of the season); (iii) seed N, P, and K uptake or tissue concentration at maturity; (iv) stover (leaf + stem + petiole + pod wall) N, P, and K uptake or tissue concentration at maturity; (v) dry mass and nutrient partitioning HIs; (vi) nutrient internal efficiency; and (vii) N/P, N/K and P/K ratios derived from total nutrient uptake at harvest. The majority of data were retrieved from tables, some from equations, and a small proportion from digitized figures. Seed yield, aboveground biomass, and plant nutrient uptake are all expressed in dry basis. Units were standardized to megagrams per hectare for seed yield and stover biomass, kilograms per hectare for nutrient uptake, and grams per 100 grams for nutrient concentration.

Data Description

Relevant experimental details of country, design, and year of experimentation were retrieved from the publications. Crops grew under contrasting soil, weather, and management

conditions (Table 1), causing large variation in all traits under study (Fig. 1). Our analysis involved three steps. First, we tested for time trends in each of the traits. Second, the database was divided into three temporal groups: 1922 to 1996 (Era I, $n = 43$), 1997 to 2006 (Era II, $n = 110$), and 2007 to 2015 (Era III, $n = 216$); the increase in data points (means indicated in parenthesis) with time reflects the increased research effort. The criteria to define the eras were: Era I spans from the first published study (Borst and Thatcher, 1931) to the commercial release of the first transgenic soybean in 1996 (Fernandez-Cornejo and McBride, 2002), Era II spans from the first transgenic soybean to >50% of the global soybean crop represented by transgenic soybean in 2006 (James, 2006), and Era III spans from this point to the present time. Average study year for each era was 1979, 2003, and 2011 (Supplemental Fig. S1). Therefore, our era-based evaluation was primarily focused in the 1980s, 2000s, and 2010s. Third, we assessed changes over time of all traits of study. The departure of each trait relative to the mean of each decade was calculated and used to graphically synthesize the main findings.

Descriptive and Statistical Analysis

Traits were plotted against time to quantify trends (Fig. 1). For each variable, a least squares linear regression was conducted, and the slope was tested for significance (Motulsky and Christopoulos, 2003). Histograms were constructed for seed yield and boxplots for N, P, and K uptake (Fig. 2).

Zero-intercept linear regression was used to derive nutrient ratios (Fig. 3A–3C) and to relate seed yield and total N, P, and K uptake (Fig. 4A–4C). An F -test was performed to compare slopes among eras using $p = 0.05$ threshold in all cases. Allometric analyses were performed to quantify the changes, primarily in the slope, of the nutrient internal efficiencies (IEs) and nutrient ratios among eras. To account for error in both x and y , a Model II regression was used (reduced major axis; Niklas, 1994) in Fig. 3A to 3C and 4A to 4C. The SMATR package (Warton et al., 2015) from the R program (R Core Team, 2017) was used to test for common slopes. Envelopes portraying maximum and minimum boundaries (0.99 and 0.01 quantile lines, respectively) for yield-to-nutrient content and nutrient ratios were calculated using the “quantreg” package in R software (Koenker, 2005, 2017). The upper boundary of the envelope represents the maximum nutrient dilution, and the lower boundary represents the maximum nutrient accumulation (Janssen et al., 1990; Witt et al., 1999). Residuals of the adjusted functions were plotted against stover, seed nutrient concentration, and nutrient HI to account for physiologically important sources of variation in these traits. The proportion of variance (R^2) was determined between residuals of the seed yield to plant traits such as plant nutrient uptake, nutrient concentration, and nutrient HI. The same procedure was implemented by Sadras (2006), Ciampitti and Vyn (2013), and Tamagno et al. (2017).

To summarize the changes in yield, nutrient uptake, and partitioning, we used the framework developed by Sadras et

Table 1. Global soybean database information, including author, year of publication, state or country, year(s) of the study, number of observations collected, and main characteristics evaluated.

Author(s)	Year of publication	State or country	Year(s) of the study	n	Factors evaluated
Borst et al.	1931	Ohio, USA	1922–1927	6	Variety, seeding rate, and planting date.
Hammond et al.	1951	Iowa, USA	1946	2	Soil fertility levels
Bataglia et al.	1976	Goiás, Brazil	1973–1976	1	P, K, and S fertilization
Larcher et al.	1984	Casamace, Senegal	1978–1979	2	Fertilization
Terman	1977	Alabama, USA	1973–1974	2	N and K fertilization
Desoky	1996	Assiut, Egypt	1995–1996	16	Variety, plant density, and N, P, and K fertilization
Vasilas	1984	Minnesota, USA	Not reported	8	N fertilization
Henderson et al.	1970	North Carolina, USA	1966–1968	3	Nutrient uptake
Bataglia et al.	1977	Sao Paulo, Brazil	1974–1976	3	Nutrient uptake
Heard	2006	Manitoba, Canada	2005	1	Nutrient uptake
Rao and Lakshmi	2009	Telangana, India	2003	18	Compost
Rathore et al.	2008	Uttarakhand, India	2006	7	Seaweed fertilizer
Najar et al.	2011	Maharashtra, India	2004	10	Fertilization
Rao et al.	2012	Karnataka, India	2005–2007	2	Fertilization
Chandel	2011	Maharashtra, India	2009	8	Fertilization
Meena and Biswas	2013	New Delhi, India	2010	8	Fertilization
Patel	2011	Madhya Pradesh, India	2009–2010	12	Fertilization
Kurihara et al.	2013	Mato Grosso do Sul, Brazil	2001	1	High yield management
Fageria et al.	2013	Tocantins, Brazil		5	Liming
Bender et al.	2015	Illinois, USA	2012–2013	2	Nutrient uptake
Tamagno et al.	2017	Santa Fe, Argentina	2009–2013	68	Survey of soybean under rainfed conditions without nutrient limitations across several fields
		Kansas, USA	2014	45	Row spacing, inoculation, plant density, and fertilization strategy
		Indiana, USA	2011–2012	54	Variety, biomass, and nutrient uptake and partitioning
Balboa and Ciampitti	Unpublished	Kansas, USA	2015	30	Row spacing, inoculation, plant density, and fertilization strategy

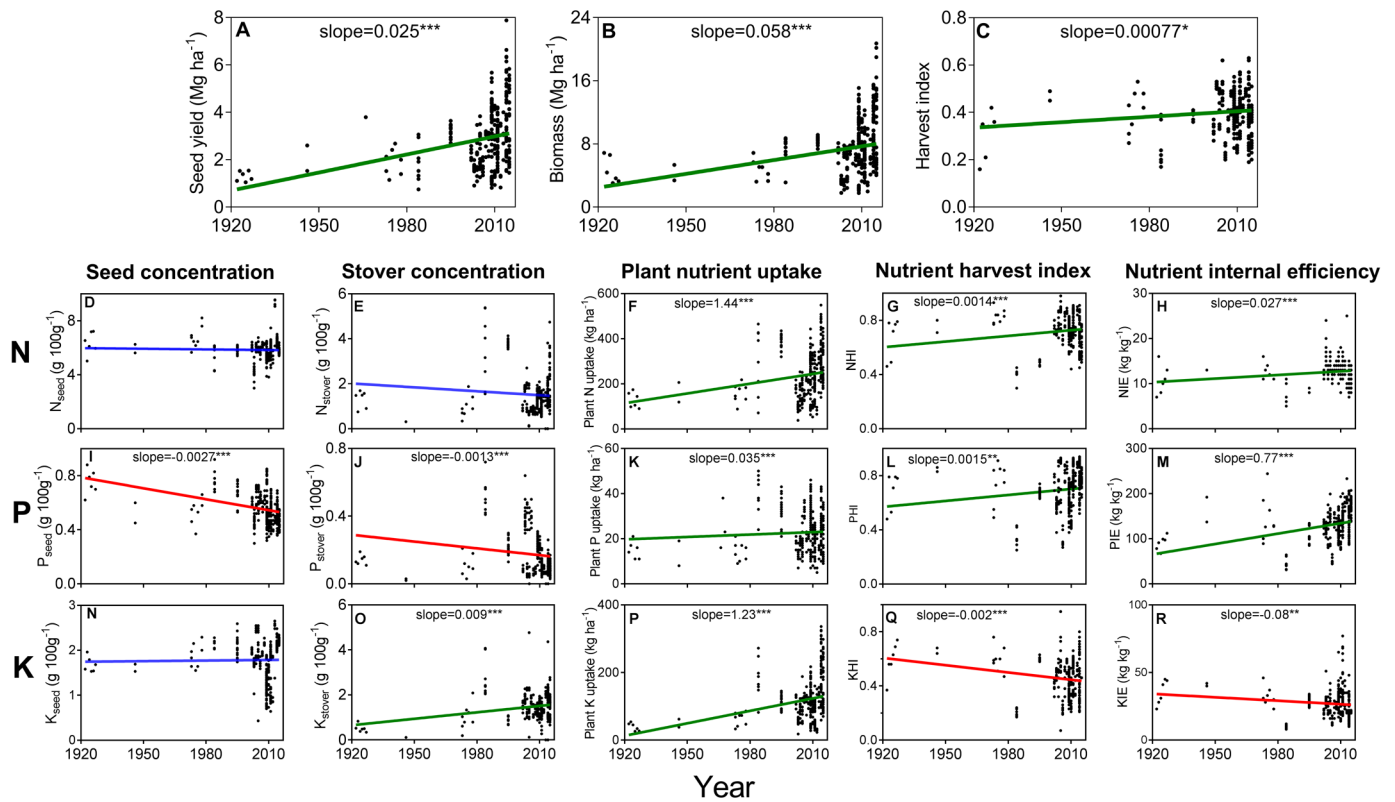


Fig. 1. (A) Soybean seed yield; (B) biomass; (C) harvest index; (D, I, and N) seed N, P, and K concentration; (E, J, and O) stover N, P, and K concentration; (F, K, and P) plant N, P, and K nutrient uptake; (G, L, and Q) N, P, and K harvest index; and (H, M, and R) nutrient internal efficiency across years for the pool data ($n = 322$). Solid lines represent linear regression. Green represents positive slope ($***p < 0.01$), red represents negative slope ($***p < 0.001$), and blue a slope not different from zero. NHI, N harvest index; NIE, N internal efficiency; KHI, K harvest index; KIE, K internal efficiency; PHI, P harvest index; PIE, P internal efficiency.

al. (2016). First, we compared the rate of change in seed yield and the rate of change in nutrient uptake in relation to the $y = x$ line, which corresponds to IE (Fig. 5A). The variation of the trait (seed yield or nutrient uptake) was calculated as a coefficient relative to the observations within an era. For example, the change in yield for the first era was determined by dividing each observation within an era by the average yield for the last year of the era, and then all observations were averaged, and the deviation from a unit was calculated (average minus one).

The rationale is that IE would increase if the rate of gain in seed yield is larger than the rate of increase in nutrient uptake (points above $y = x$) whereas the reverse would indicate decreased IE (i.e., points below $y = x$). If the value is different than zero, then the plant trait evaluated for that era presented a change compared with the last year within the era. The slope of the relation of yield to nutrient uptake represents the IE that can be mathematically expressed, as proposed by Sadras (2006), as:

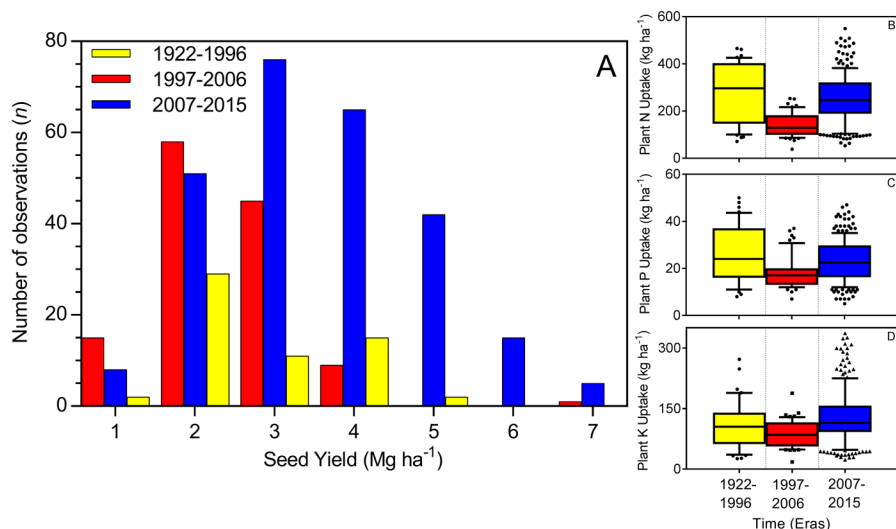


Fig. 2. (A) Soybean seed yield by era and plant N, P, and K uptake (B, C, and D, respectively) by era.

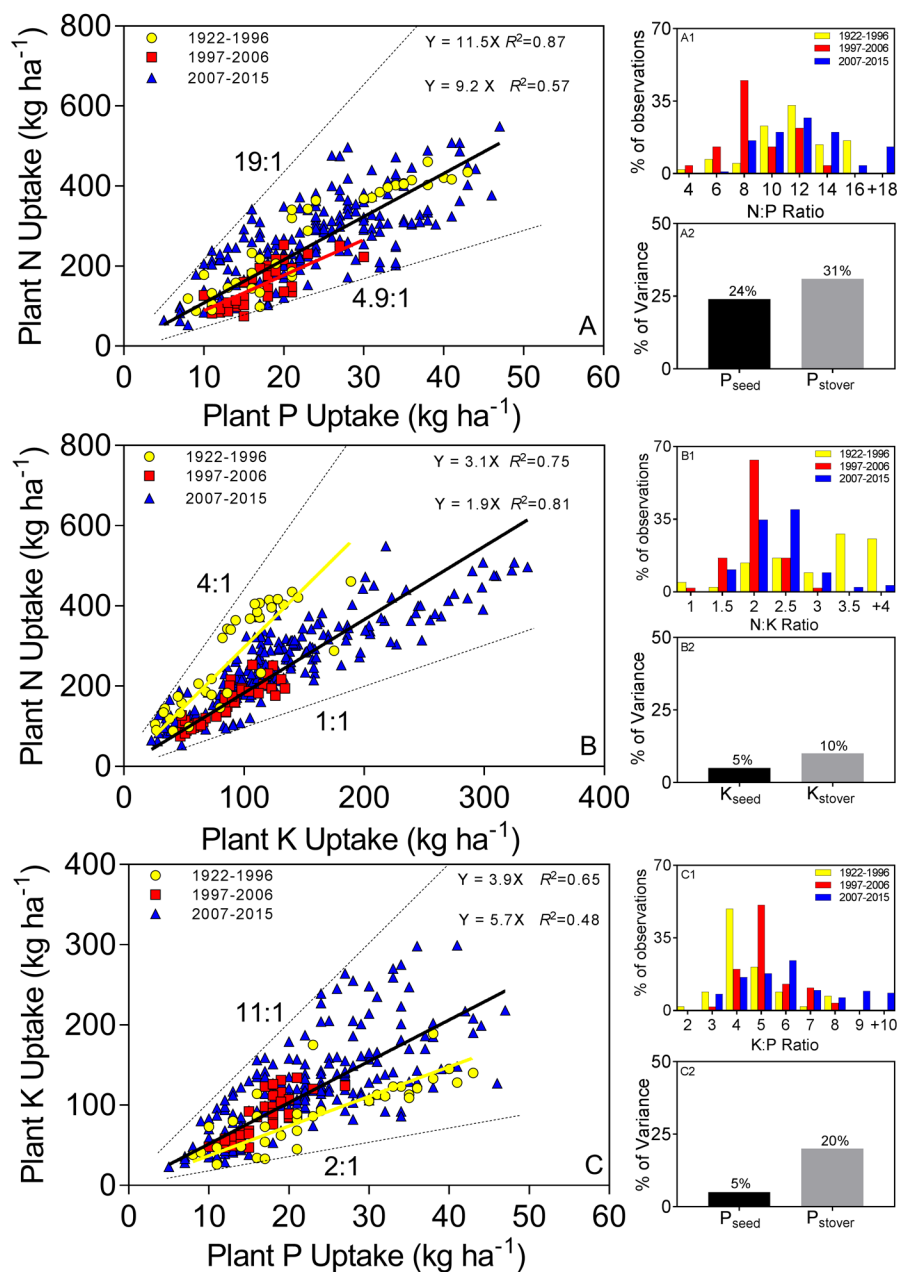


Fig. 3.(A) Plant N uptake as a function of plant P uptake, (B) plant N uptake as a function of K uptake, and (C) plant K uptake as a function of plant P uptake. N/P, N/K, and P/K ratio by era (A1, B1, and C1, respectively). Percentage of variance (R^2) explained by the linear regression between (A2) residuals of Fig. 3A as a function of seed and stover P concentration, (B2) residuals of Fig. 3B as a function of seed and stover K concentration, and (C2) residuals of Fig. 3C as a function of seed and stover P concentration. Dotted lines indicate boundaries for maximum and minimum ratio for each dataset. Due to lack of significant difference ($p > 0.05$) in slopes, Eras I and II were pooled in Fig. 3A (black line); Eras II and III were pooled in Fig. 3B and 3C (black line). Linear function for pooled dataset: (A) $Y = 11.5X$ ($R^2 = 0.87$), (B) $Y = 1.9X$ ($R^2 = 0.81$), (C) $Y = 5.7X$ ($R^2 = 0.48$).

$$IE = \frac{Y}{\text{Nutrient}} = \left(\frac{Y}{\text{Nutrient}_{\text{seed}}} \right) \left(\frac{\text{Nutrient}_{\text{seed}}}{\text{Nutrient}_{\text{seed}} + \text{Nutrient}_{\text{stover}}} \right) \quad [1]$$

where Y is grain yield, “Nutrient” is plant nutrient uptake, “Nutrient_{seed}” and “Nutrient_{stover}” are nutrient concentration in seed and stover. The second term of Eq. [1] represents nutrient harvest index:

$$\text{Nutrient HI} = \frac{\text{Seed nutrient}}{\text{Seed nutrient} + \text{Stover nutrient}} \quad [2]$$

where “Seed nutrient” refers to seed nutrient uptake and “Stover nutrient” to stover nutrient uptake.

Second, to analyze the contribution of nutrient seed concentration and nutrient harvest index to the variation in IE, the rate of change in nutrient seed concentration was plotted against the rate of change in yield per unit nutrient uptake using the $y = -x$ line as reference (Fig. 5B). Points aligned around $y = -x$ indicate that a given increase (reduction) in yield per unit nutrient uptake would result in a proportional reduction (increase) in seed nutrient concentration unless nutrient HI changes.

RESULTS

Time Trends for Yield and Nutrient Traits

Seed yield, biomass, and HI showed positive trends across years (Fig. 1A–1C). The rate of increase in biomass, seed yield, and HI was $0.025 \text{ Mg ha}^{-1} \text{ yr}^{-1}$, $0.058 \text{ Mg ha}^{-1} \text{ yr}^{-1}$, and 0.0008 yr^{-1} , respectively. Seed yield ranged from 1.3 Mg ha^{-1} in the 1930s to 3.2 Mg ha^{-1} in the 2010s, with a minimum of 0.83 Mg ha^{-1} and a maximum of 7.88 Mg ha^{-1} (Fig. 1A). Total N, P, and K uptake showed a positive time trend for the pooled data (Fig. 1F, 1K, and 1P). The rate of increase in nutrient uptake was $1.57 \text{ kg N yr}^{-1}$, $0.076 \text{ kg P yr}^{-1}$, and $0.51 \text{ kg K yr}^{-1}$ (Supplemental Table S1).

Seed N (N_{seed}) and K (K_{seed}) concentrations were stable across years (Fig. 1D and 1N). The N_{seed} displayed the smallest variation across years, with a CV of 14% (Supplemental Table S2). Phosphorous seed concentration declined with time (Fig. 1I), with an average rate of $-0.0027 \text{ g } 100 \text{ g}^{-1} \text{ yr}^{-1}$. Stover nutrient concentration remained stable for N (Fig. 1E), decreased for P at $-0.0013 \text{ g } 100 \text{ g}^{-1} \text{ yr}^{-1}$ (Fig. 1J), and increased for K at $0.009 \text{ g } 100 \text{ g}^{-1} \text{ yr}^{-1}$ (Fig. 1O). For all three nutrients, concentration varied more in stover than in seed.

Nitrogen HI increased at a rate of 0.0014 yr^{-1} (Fig. 1G), starting from as low as 0.66 in 1930 and ending with an average of 0.72 in 2010. Similarly, HI for P (PHI) increased with time (Fig. 1L) at 0.0015 yr^{-1} , from 0.68

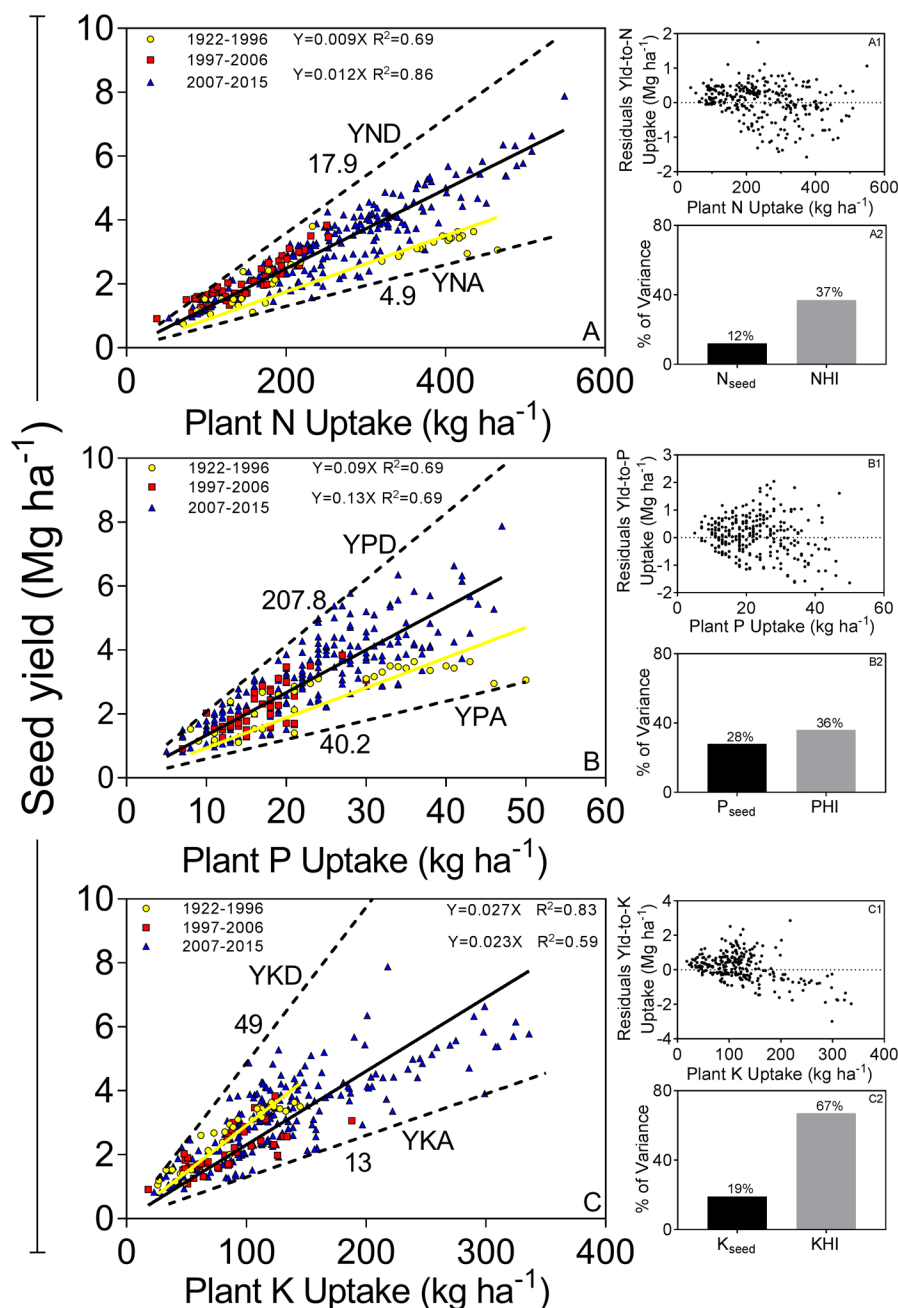


Fig. 4. Seed yield as a function of (A) plant N uptake, (B) plant P uptake, and (C) plant K uptake. (A1), (B1), and (C1) show residuals for the fitted functions in (A), (B), and (C), respectively. (A2) Proportion of variation (R^2) provided by the linear regression between residuals of Fig. 4A as a function of seed N concentration and N harvest index (NHI); (B2) Residuals of Fig. 4B as a function of seed P concentration and P harvest index (PHI); and (A3) residuals of Fig. 4C as a function of seed K concentration and K harvest index (KHI). YNA, yield N accumulation; YND, yield N dilution; YKA, yield K accumulation; YKD, yield K dilution; YPA, yield P accumulation; YPD, yield P dilution. Due to lack of significant difference ($p > 0.05$) in slopes, Eras I and II were pooled in Fig. 3A to 3C (black line) Linear function for pooled dataset (A) $Y = 0.012X$ ($R^2 = 0.86$), (B) $Y = 0.13X$ ($R^2 = 0.69$), (C) $Y = 0.023X$ ($R^2 = 0.59$).

to 0.73 between 1930 and 2010. Potassium HI (KHI) declined over time (Fig. 1Q) at -0.018 yr^{-1} , from 0.59 to 0.43 between 1930 and 2010. Nutrient internal efficiency increased over time for both N at $0.026 \text{ kg seed kg}^{-1} \text{ N yr}^{-1}$ and P at $0.767 \text{ kg seed kg}^{-1} \text{ P yr}^{-1}$ (Fig. 1H and 1M);

IE for K (KIE) declined at $-0.085 \text{ kg seed kg}^{-1} \text{ K yr}^{-1}$ (Fig. 1R).

The largest proportion of the change in IE for N (NIE) was related to yield improvement, followed by increases in HI for N (NHI) and N uptake with stable N_{seed} and stover N concentration (N_{stover}) (Fig. 1). The change in IE for P (PIE) was related to an increase in yield, followed by improvement in PHI and a decrease of seed P concentration (P_{seed}) (Fig. 1I–1M). The largest proportion of change in KIE was related to a larger increase in yield relative to total plant K uptake. Potassium HI also decreased with a slightly improvement in K_{stover} over time (Fig. 1O and 1Q).

Trends by Eras

Yield and total uptake of N, P, and K for each era are presented in Fig. 2. As documented in Fig. 1, average seed yield and nutrient uptake increased across historical eras (Supplemental Table S2).

Nutrient Stoichiometry

Figures 3A to 3C show the change in nutrient ratio with era. Nitrogen/phosphorus ratio ranged from 4.9 to 19.0, with greater values for Eras I and III than for Era II. The slope for Eras I and III was larger (11.5) than for Era II (9.0). This is related to the smaller total nutrient content (N, P, and K) for Era II compared with the other eras. The histogram shows higher frequency of greater N/P ratio for Eras I and III than for Era II (Fig. 3A1). Overall, P_{seed} and stover P concentration (P_{stover}) accounted for $>50\%$ of the N/P variation (Fig. 3A2).

Nitrogen/potassium ratio ranged from 1 to 4 (Fig. 3B). Eras II and III clustered and have smaller ratios than Era I. Era I showed higher frequency of higher N/K ratio, followed by Eras III and II (Fig. 3B1). For Era I, N/K averaged 3.1 and decreased to 1.9 for Eras II and III. Seed and stover K concentrations accounted for 5 and 10% of the variation in the residuals of the N/K ratio.

Potassium/phosphorus ratio ranged from 2 to 11 (Fig. 3C). Era I showed greater frequency of smaller K/P ratios (Fig. 3C1). Analysis of slopes clustered Eras II and III, with smaller ratios for Era I. The P_{stover} explained 20% of the variation in the K/P ratio and P_{seed} only contributed

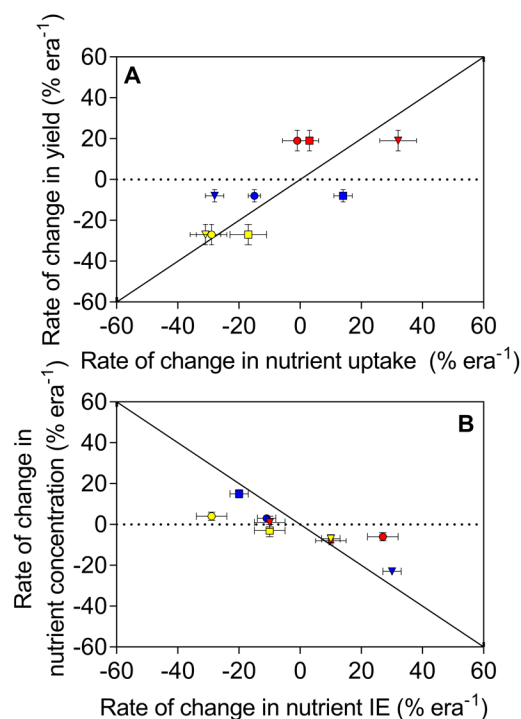


Fig. 5. (A) Rate of change of soybean seed yield as a function of rate of change of nutrient uptake, (B) rate of change of nutrient concentration. Symbols indicate nutrients: circle = N, square = P, inverted triangle = K. Each point has standard error bars for X and Y variables. Rate of change for a trait was calculated for each era considering the average value for the trait during the last year of the era as a reference. Color indicates era: yellow = Era I, red = Era II, blue = Era III. IE, internal efficiency.

to 5% of the variation (Fig. 3C2). Stover P concentration accounted for a larger proportion of the variation in K/P relative to P_{seed} , in agreement with the larger decrease in P_{stover} (36%) relative to P_{seed} (19%) among eras (Supplemental Table S2).

Nutrient Internal Efficiency

For both N and P, IE clustered for Eras II and III and was superior to the IE for Era I. For K, IE clustered for Eras II and III and was inferior to Era I. Average NIE increased 33% (Fig. 4A) from Era I (9 kg seed kg⁻¹ N) to Eras II and III (12 kg seed kg⁻¹ N); average PIE (Fig. 4B) increased by 44% from Era I to Eras II and III (90 to 130 kg seed kg⁻¹ P ha⁻¹); KIE decreased from 27 to 23 kg seed kg⁻¹ K from Era I to Eras II and III (Fig. 4C). Variation in seed yield per unit of nutrient uptake was portrayed with two boundaries (dotted lines in Fig. 4A–4C).

Residuals for the relationship between seed yield and nutrient uptake are presented in Fig. 4A1 to 4C1. The residuals were regressed against nutrient concentrations in seed and in stover to further dissect the nutrient IEs. The proportion of variance (R^2) explained by each trait is presented in Fig. 4A2 to 4C2.

Nitrogen HI accounted for 37% of the variation in NIE, whereas N_{seed} accounted for 12% of the NIE variation.

Phosphorous HI accounted for 36% of the variation in PIE, and 28% was explained by P_{seed} . Potassium HI accounted for 67% of the variation in KIE, with a smaller contribution from K_{seed} (19%). For all three nutrients, nutrient HI accounted for a large proportion of the variation in the IE, also reflected in the lesser variation relative to NHI. In contrast, N_{stover} and P_{stover} decreased 9.5 and 15.5% for Eras II and III relative to Era I, respectively, and K_{stover} increased by 37% comparing Era I with the average of Eras II and III (Fig. 1E, 1J, and 1O; Supplemental Table S2).

Relative Traits Change

Nitrogen IE improved between Eras I and III, as improvements in yield were unmatched by increased N uptake (Fig. 5A). The relative variation in N_{seed} (−1%, y-axis, Fig. 5B) was small compared with variation in NIE (+30%, slope, Fig. 5A) for Eras I and III. Changes in nutrient seed concentration and nutrient HI drive changes in IE (Eq. [1]). Since N_{seed} remained approximately stable (Fig. 5B), variations in IE (Fig. 5A) were driven by nutrient HI. The increase in NIE was partially a consequence of the increase in NHI (+19%, Fig. 1G). Similar to N, PIE increased between Eras I and III, as yield gain was not matched by increase in P uptake (Fig. 5A). The increase in PIE (+39%) was more closely related to increase in PHI (+14%, Fig. 1L) compared with P_{seed} (−19%, Fig. 1I and 5B).

For K, the relative improvement in uptake was larger (+35%, Fig. 5A) than the increase in K_{seed} (Fig. 5B), evidenced as a major distance between triangles in the x-axis compared with variations in y-axis. Reduction in KIE (−4%) was related to a time trend increase in K_{stover} (+28%, Fig. 1O) with a concomitant decrease in KHI (−11%, Fig. 1Q).

In summary, improvement in nutrient IE (Fig. 5A) was more proportionally explained by changes in nutrient HIs, with more dispersion from the 1:1 line, whereas nutrient seed concentrations remained stable, primarily for N and K (Fig. 5B). Therefore, as previously stated, maintenance of nutrient seed concentrations for N and K were obtained by compensatory increases in nutrient HI, with an insufficient change for P. Phosphorus seed concentration declined over time, more proportionally than the increase in PHI (Fig. 1I and 1L).

DISCUSSION

Yield, Biomass, and Harvest Index

Our database, primarily from small-plot experiments, showed similar yield gain per year to the 25.8 kg ha⁻¹ yr⁻¹ reported by FAO between 1960 and 2014 (FAO, 2017). Historical time trends for soybean yield average increase have been documented in the United States (23 kg ha⁻¹ yr⁻¹; Long, 2013; Rowntree et al., 2013; Specht et al., 2014; Wilson et al., 2014), China (11 kg ha⁻¹ yr⁻¹; Wu et al., 2015), Argentina (44.33 kg ha⁻¹ yr⁻¹; de Felipe et al., 2016), and Brazil (41 kg ha⁻¹ yr⁻¹; de Toledo, 1990). Our database

indicates that yield improvement was mostly associated with increased biomass and little or no change in HI, in agreement with previous studies (Schapaugh and Wilcox, 1980; Spaeth et al., 1984; Johnson and Major, 1986).

Nutrient Ratios

Nutrient ratios can help to predict nutrient limitations better than individual nutrient concentrations (Güsewell, 2004; Sadras, 2006; Malingreau et al., 2012; Divito and Sadras, 2014; Koerselman and Meuleman, 2017). Nutrient ratios are sensitive to changes in biomass (Greenwood et al., 2008; Ciampitti and Vyn, 2014). Nutrient availability and nutrient uptake for immediate use and storage are major sources of variation in nutrient ratio (Bollons and Barraclough, 1999; Ågren, 2008). Liebig's law inadequately assumes a single limiting factor at a given time (Sinclair and Park, 1993), whereas in reality, crop growth is often colimited by multiple factors (Sadras, 2005; Ågren et al., 2012; Mooney et al., 2012).

In this review, average N/P ratio for modern soybean varieties was 11.5, as compared with the range of 12 to 13 that is typical across terrestrial plants (Güsewell and Koerselman, 2002; Güsewell, 2004; Knecht and Göransson, 2004). Nitrogen/phosphorous ratio averaged 4.5 for oilseed crops, 5.6 for cereals, and 8.7 for legumes (Sadras, 2006). Nitrogen/phosphorous ratio has been used mainly to assess whether N or P is more limiting for biomass production. Tamagno et al. (2017) reported a N/P ratio of 11.4 for soybean with experiments in Argentina and the United States. In legumes, Sadras (2006) found that differences in P rather than N concentration better explain the N/P ratio variation, since N is more closely regulated by the plant compared with P. Nitrogen concentration in seeds is a conservative trait—small variation (Sinclair and de Wit, 1976; Jin et al., 2011) with a neutral trend over time (Long, 2013). Variations between N/K and K/P reported in this review were greatly influenced by the increase in K uptake in Era III.

Nutrient Internal Efficiency

In previous decades, scientific research improved the understanding of nutrient uptake and underlying genetics, but less effort was directed towards comprehension of nutrient IE (Santa-María et al., 2015). In agreement with our findings, analysis of modern soybean varieties showed larger changes in IE than in total nutrient uptake (Bender et al., 2015; Gaspar et al., 2017; Yang et al., 2017). Selection for yield increased NIE and PIE but reduced KIE. Soybean seed requires large amounts of N per unit of C relative to other crops (Sinclair and de Wit, 1975), and superior N demand is accompanied by superior P and K requirements (Bender et al., 2015; Gaspar et al., 2017; Yang et al., 2017). Recently, Cafaro La Menza et al. (2017), reporting fertilizer N response in soybeans grown

in high-yielding environments ($>6 \text{ Mg ha}^{-1}$), found that indigenous N supply was insufficient to fulfill N requirement with lower N_{seed} when no N was applied. Similar findings were also reported by Tamagno and Ciampitti (2017) but in lower-yield environments (average 3.5 Mg ha^{-1}), with superior soybean yield to N fertilization explained by longer duration of the seed-filling period, even when biological N fixation was partially inhibited.

The conserved N_{seed} documented in this review is in contrast with other studies reporting negative time trends (Cregan and Yaklich, 1986; Wilcox, 2001; Long, 2013). Nonetheless, our dataset provides a robust explanation, whereby N_{seed} was conserved because both N uptake and NHI increased to fully compensate for the increase in yield.

Phosphorous IE increased over time as a consequence of a reduction in P_{seed} (with reduction in the nutrient partitioning to the seed), and also in P_{stover} , with similar total P uptake across eras. In general, there are two mechanisms to modify PIE within the plant: (i) increasing remobilization from storage, or (ii) reducing the partitioning to developing reproductive organs. For field crops, P_{seed} decreased over time with breeding in association with improvement in HI (Veneklaas et al., 2012).

Variation in KIE can be achieved by (i) substitution of K with other cations, (ii) K translocation between organelles, cells, and organs to regulate K concentration in the cytoplasm, and (iii) K translocation to seed (Gerloff, 1987; Sattelmacher et al., 1994; Rengel and Damon, 2008). In this review, K_{stover} was larger in Eras II and III than in Era I, which can be related to the increase in K fertilization since 1960 (USDA-ERS, 2016). Several authors have reported increases in K_{stover} with K fertilization and without yield response in corn and soybean (Heckman and Kamprath, 1995; Randall et al., 1997; Clover and Mallarino, 2013). Luxury nutrient uptake occurs when uptake is above the crop requirement for a target yield level. As presented by Ciampitti and Vyn (2012), luxury nutrient consumption also occurs if soil nutrient status is high and crop nutrient removal is greater than normal, promoting unbalanced plant nutrient ratios.

Changes in Nutrient Internal Efficiency: Dissecting Critical Components

Changes in nutrient IEs were attained differently for each nutrient. Nitrogen HI increased over time (1922–2015), as reported by others (Jin et al., 2011; Long, 2013), but with N_{stover} remaining stable (Long, 2013). Superior PIE was achieved by a decrease in both P_{seed} and P_{stover} over time. Larger yields were associated with improvement in K uptake, reductions in K_{seed} and KHI, and a substantial increment in K_{stover} , leading to a reduction in KIE over time. Increases in K_{stover} were also reported by others (Heckman and Kamprath, 1995; Randall et al., 1997; Clover and Mallarino, 2013).

Several studies (Vitousek et al., 2009; Peñuelas et al., 2013; Bouwman et al., 2017) reported nutrient imbalances in croplands around the world, but improving nutrient IEs may contribute to reversing this situation.

CONCLUSIONS

Historical analysis of soybean for the period 1922 to 2015 showed that (i) seed yield improved from 1.3 Mg ha⁻¹ in 1930 to 3.2 Mg ha⁻¹ in 2010; (ii) seed yield increase was primarily driven by higher biomass rather than HI; (iii) NHI and PHI both increased over time; (iv) seed nutrient concentration remained stable for N but declined for both P (18%) and K (13%); (v) stover nutrient concentration remained stable for N, declined for P, and increased for K; (vi) nutrient ratios show different trends for N/P (Era I and III > II), N/K (Era I > II and III), and K/P (Era II and III > I); (vii) nutrient IE increased for N (33%) and P (44%) but decreased for K (11%); and (viii) variations in nutrient IEs were primarily explained by changes in nutrient HIs for N and K but were equally explained by both PHI and P_{seed}. A focus on plant nutrient ratios and their relations to crop growth rate is likely to provide better tools for nutrient management.

Conflict of Interest

The authors declare that there is no conflict of interest.

Supplemental Material Available

Supplemental material for this article is available online.

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